Standard error + X versus X̄

1 RECOGNISE IT

- Compare one observation with an average of n observations.
- Asked for the spread of sample means or effect of larger n.
- Population σ is known.

2 DECISION / METHOD

1. Name the random object: X or X̄.
2. X uses centre μ and SD σ.
3. X̄ uses centre μ and SE σ/√n.
4. Larger n narrows X̄, not the population itself.

3 FORMULA

$SD(X)=\sigma$
$SE(\bar{X})=\sigma/\sqrt{n}$
$Var(\bar{X})=\sigma^2/n$
$E(\bar{X})=\mu$

4 FORMULA SHEET VS WHAT I KNOW

Gives: SE(X̄)=σ/√n.
Still know: SD describes individuals; SE describes statistic-to-statistic variability.

5 EXCEL ROUTE

- `=sigma/SQRT(n)`
- raw sample: `=AVERAGE(range)`
- never use STDEV.S when population σ is explicitly supplied.

6 TINY WORKED EXAMPLE

If σ=18 and n=81, `SE(Xbar)=18/SQRT(81)=2`. Individuals still have SD 18.

7 TOP EXAM TRAPS

- Using σ instead of SE for X̄.
- Dividing by n instead of √n.
- Saying larger n reduces population SD.

8 SAY ALOUD

X varies with σ; sample means vary with σ divided by the square root of n.

9 EXIT CHECK

If n is multiplied by 4, what happens to SE(X̄)?

ONE-GLANCE VISUAL ROUTE

